Monday, May 18, 2026 8:44 PM

Logistic Regression ⇒

(*) Last Class ⇒
(*) New ones

(*) Agenda

→ use case
→ Log Reg^n
→ geometric intuition
→ implementation

* Telecom company ⇒ AIRTEL

(*) Airtel → 3 lD

→ CHURNing

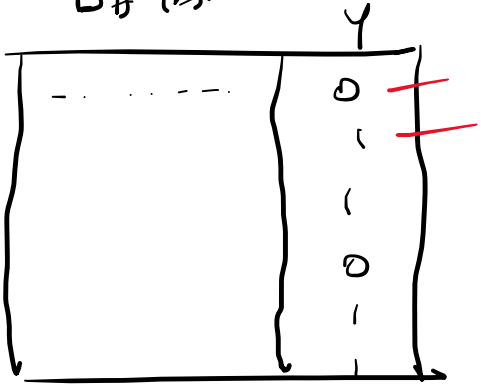
(*) CHURN Rate ⇒
||
(*) Rate at which customers are leaving the company

CHURN Prediction Problem

(x) Predictive analytics ⇒

ML → Leavers or not

(x) Customer CHURN ⇒



(+) Type of problem ⇒ Supervised

Discrete

0 or 1 → Categorical o/p

0 (NOT CHURN)

1 (CHURN)

→ Binary classification

0, 1, 2, 3

→ Multiclass

(+)

ML → find mapping b/w X & Y

→ line → w vectors
w_0

(*) $y = f(x)$ mapping func

algo → Logistic Regression ⇒

↳ × Regression

→ Classification →

→ Classifier

→ predict [0, 1]

→ Best fit line → min. ERROR
or
MSE

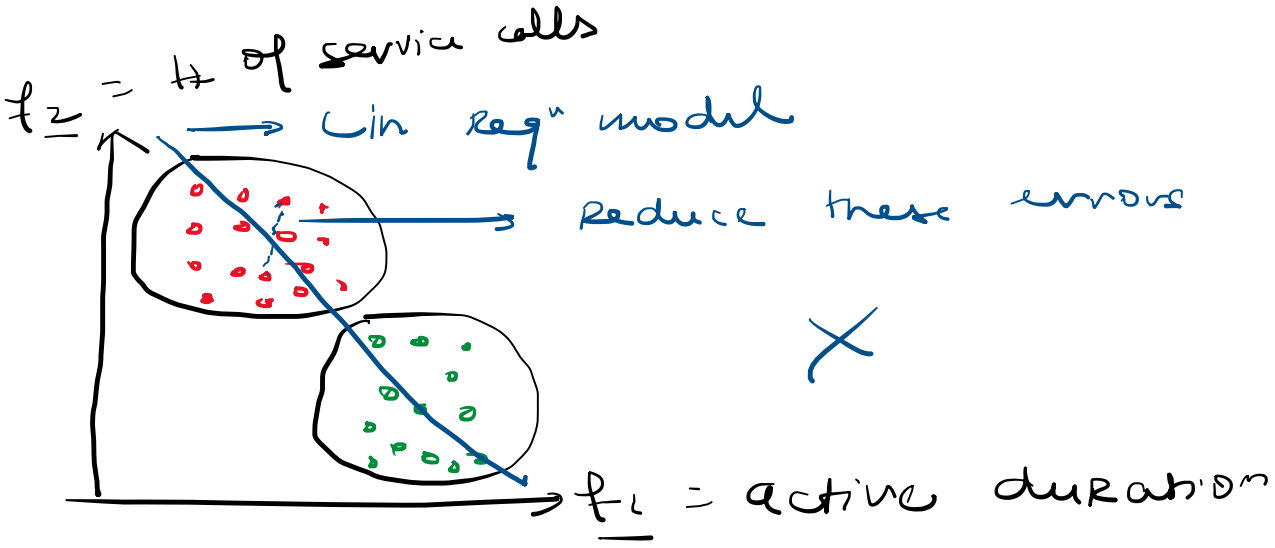
Linear Regression (*)

$\hat{y} = w^T x + w_0$

min. MSE

(*) Loss func

×



Vodafone → Vi

Goal of both algos is different

Lin Reg^n → Best fits line

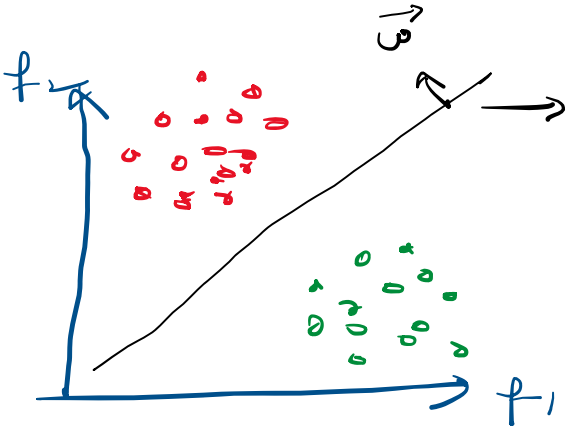
Logistic Reg^n → Best line that separates the dates

Loss func^n $\Rightarrow$ diff loss func^n

(*) Opt- eq^n $\rightarrow$ MSE

Logistic Reg^n $\Rightarrow$

line, plane, hyperplane



$\rightarrow w^T x + w_0$

seperable line $\rightarrow$ $w^T x + w_0 = 0$

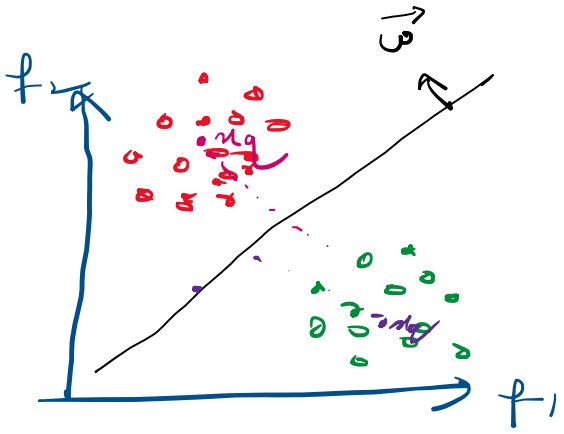
$w_1 x_1 + w_2 x_2 + w_0 = 0$

$w_1 = 2$, $w_2 = -4$, $w_0 = 1$

$$2x_1 - 4x_2 + 1 = 0$$

(7) Best line

+ve -ve



x_q → pred"

CHURN or NOT CHURN?

$w_1 x_q + w_0$

x_q → z_q

↳ Dist. from the line

(x) Half spaces

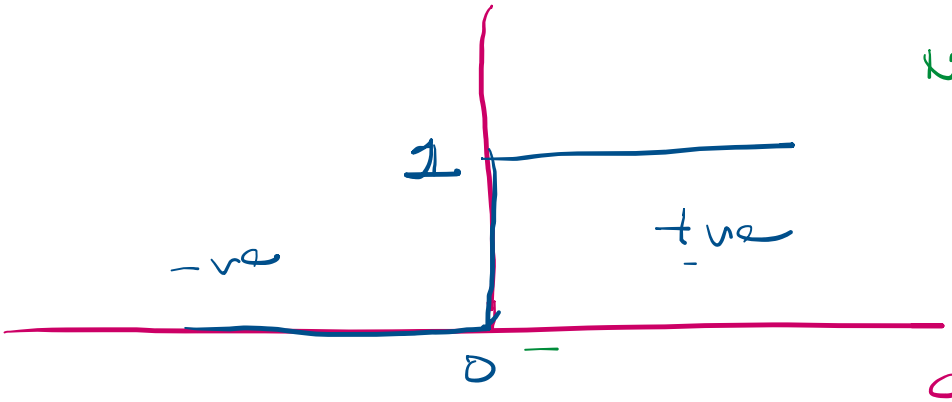
+ve → +ve HP →

-ve → -ve HP

-ve → NOT CHURN

→ mathematical function

↳ * step func^n



Not smooth

means

(x) for ML

dist. (z)

$w^T x + w_0$

0⁻ or 0⁺

(*) $g(\text{distance})$ or $g(z)$ $= \begin{cases} 1 & ; z \geq 0 \\ 0 & ; z < 0 \end{cases}$

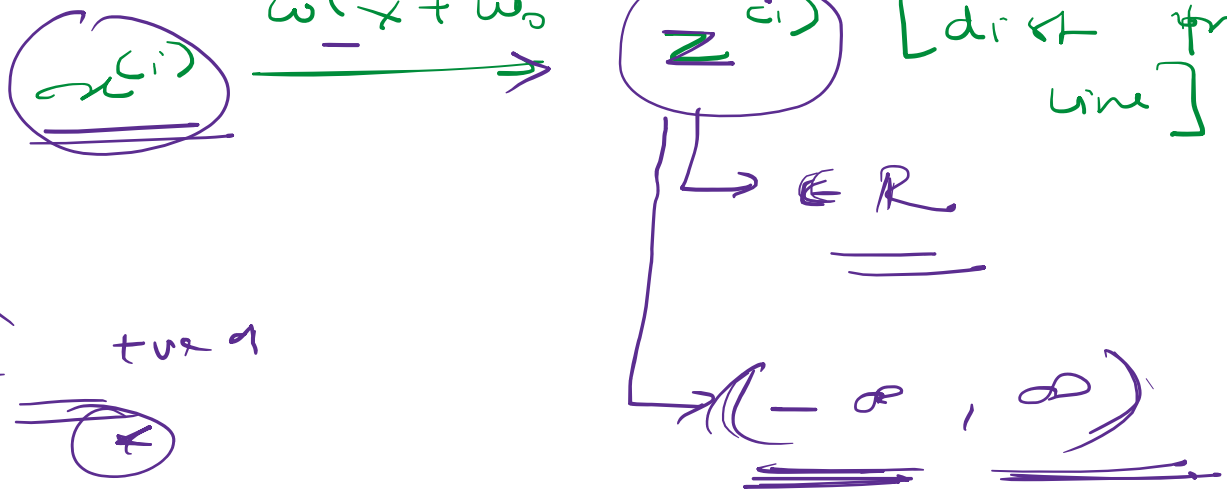
X

Problems ⇒

(a) gradients ⇒ diff^ as 0

- gradients ⇒ diff^ as 0
- we get stuck
- Sol^n ⇒
- we can change the func^

$z^{(i)} = w^T x + w_0$



- $z^{(i)}$ —func^→ 0 to 1

$z^{(i)}$ —step func^→ {0 to 1}

$\{0, 1\}$ → set

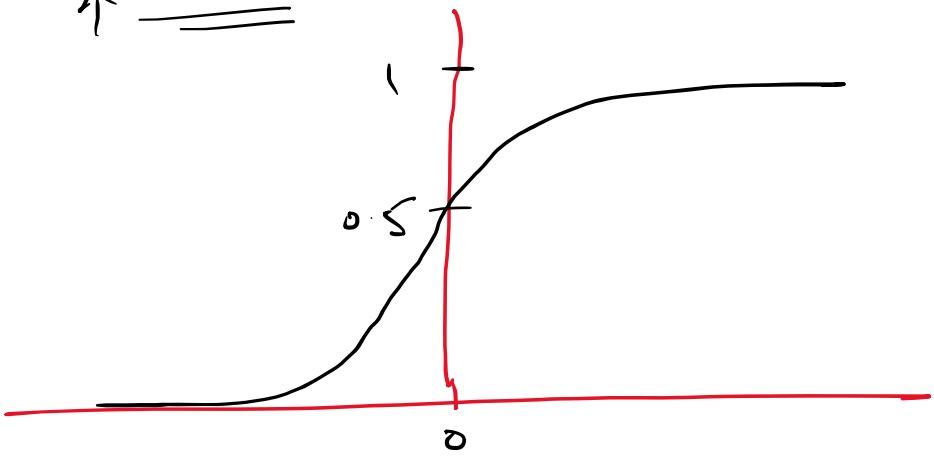
$z^{(i)}$ → $g(\,)$ → $(0, 1)$

↳ Range → value B/w 0 & 1

Smoother version

Sigmoid func^n

↑ smoother version



S – shape curve

dist $z = w^T x + w_0$

dist = +ve (value > 0.5)

dist = -ve (value < 0.5)

dist = 0 $\Rightarrow$ 0.5

$z = w_1 x + w_0$

sigmoid

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$ 0 to 1

$z \to +\infty$: $\sigma(z) = 1$

= v. large

100%.

$z \to -\infty$: $\sigma(z) = 0$

0%. chance

$z \to 0$: $\sigma(z) = 0.5$

⊛ 50-50 chance

Quiz time!

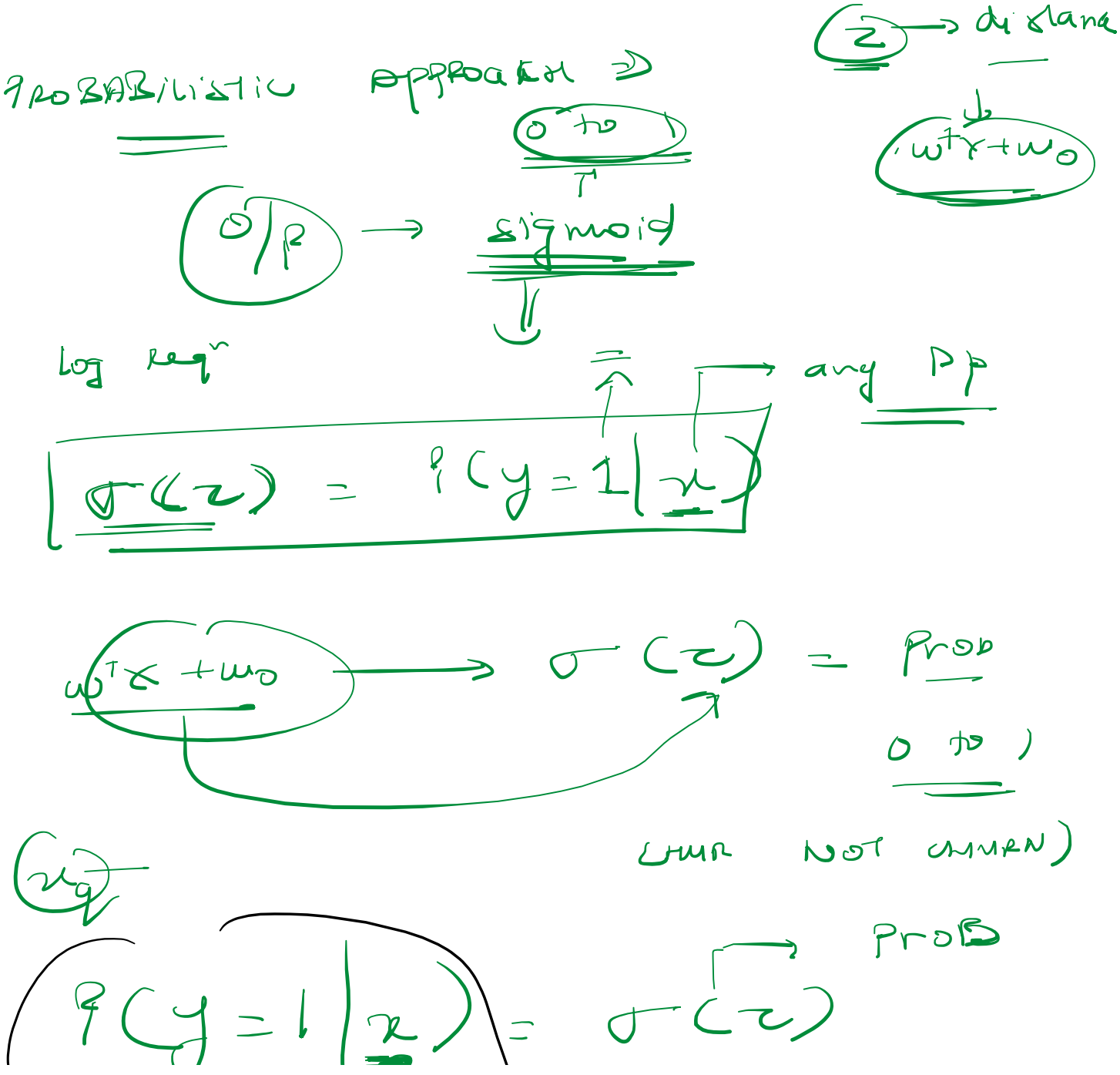
Time Left: 21s

What happens when the input to the sigmoid function is a very large negative value?

4 users have participated

A	The output becomes negative	0%
B ✓	The output approaches 0	
C	The output approaches 1	25%
D	The output becomes undefined.	25%

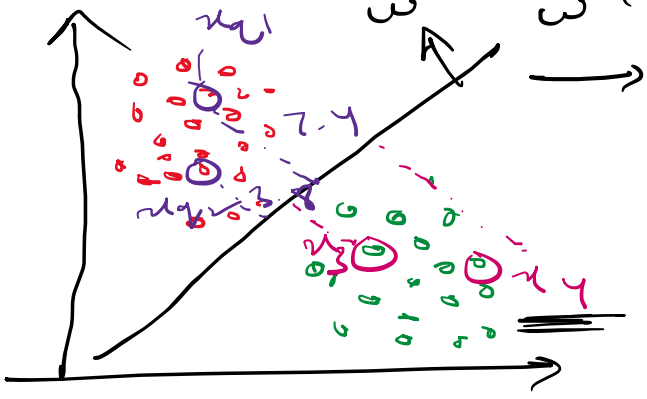
End Quiz Now



$f(y=1|x) = \sigma(z)$

dist.

$\sigma(z) =$ Prob.



$w^T x + w_0 = 0$

$\longrightarrow$ Best separator

get a obj. func^n

↓

- opt.

↓

apply GP

x_{q1} x_{q2}

x_{q1} $\Longrightarrow$

DP $\Longrightarrow$ farther $\rightarrow$ more confident

x_{q1} $(\vec{w}, w_0)$ $\rightarrow$ model x_{q2}

x_1 (w_1, w_2) x_2

$\downarrow$ line $(w^T x^1 + w_0)$ $\downarrow$ line

$z^{(1)} = 7.9$ $z^{(2)} = 8.8$

$\sigma(z_1) = 0.99$ $\sigma(z_2) = 0.76$

99% 76%

Diff

Prob.

$y \rightarrow 0, 1$

$\hat{y} \rightarrow 0 - 1$

Healthcard $\rightarrow$ 0 1 1

x_1 x_2

99% 20%

Thresholding

$\sigma(z) \geq 0.5 \rightarrow 1$

$\sigma(z) < 0.5 \rightarrow 0$

x_3 — line

$z^3 = -2.3$

$\sigma(z^3) = 0.09$

9% chance

x_4 — line

$z^4 = -5.1$

$\sigma(z^4) = 0.08$

8% chance

Smaller no.

x^4 leaving the company

$P(\text{NOT leaving} \mid x_4) = 1 - 0.09$

P(NOT leaving | x_4) = 1 − 0.5

= 61%

dist (−∞, ∞)

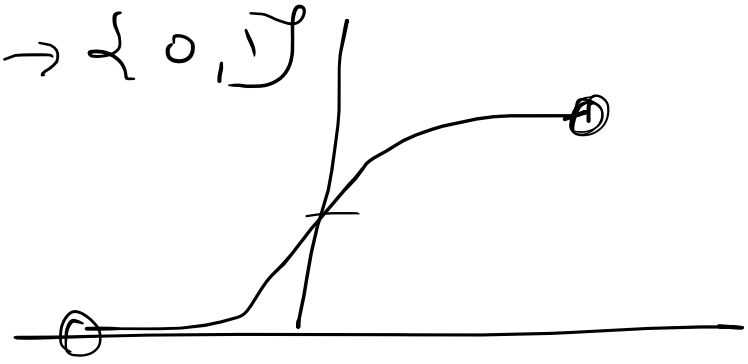
$\sigma(z)$ →

P(Y=0 | $x^{(i)}$) = 1 − 0.08 = 0.92

= 92%

$y \to \{0, 1\}$

$\hat{y} \in (0, 1)$



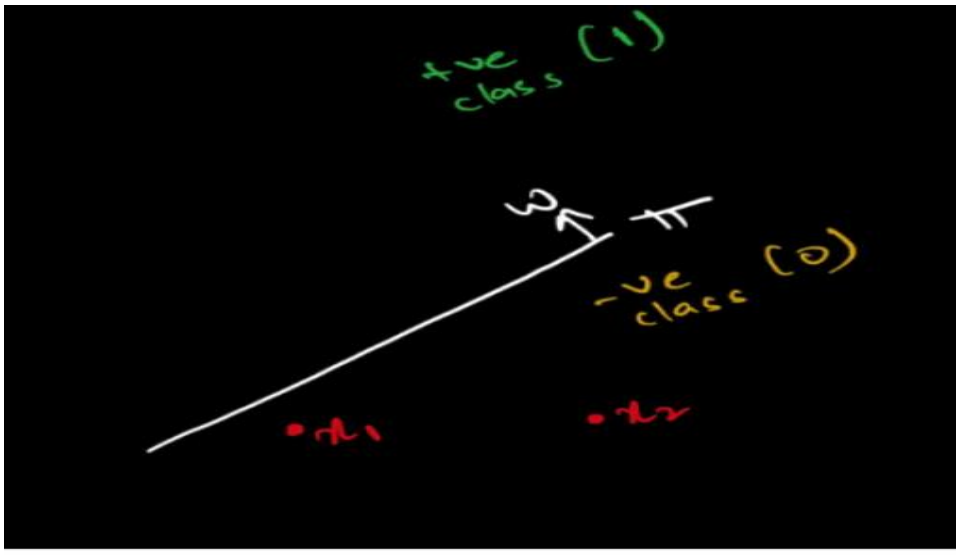
DS → ★

TRUSS PROB. MORE

Healthcare

FRAUD

★ x_q ⇒ eq → $z^{(i)}$



10:17

Initially

Which point will have a higher probability of belonging to class 1?

32 users have participated

A	x1	72%
B	x2	28%

End Quiz Now

Conclusion

Sigmoid → Smooth & diff.

squashes values (0, 1)

→ Prob of class (1)

$P(y_i = 1 | x_i)$

monotonically ↑ func.

Pred° = ?

Loss func°

$$L(y, \hat{y}) = -\frac{1}{n} \sum_{i=1}^{n} \left(y^{(i)} \cdot \log(\hat{y}^{(i)}) + (1 - y^{(i)}) \cdot \log(1 - \hat{y}^{(i)}) \right)$$

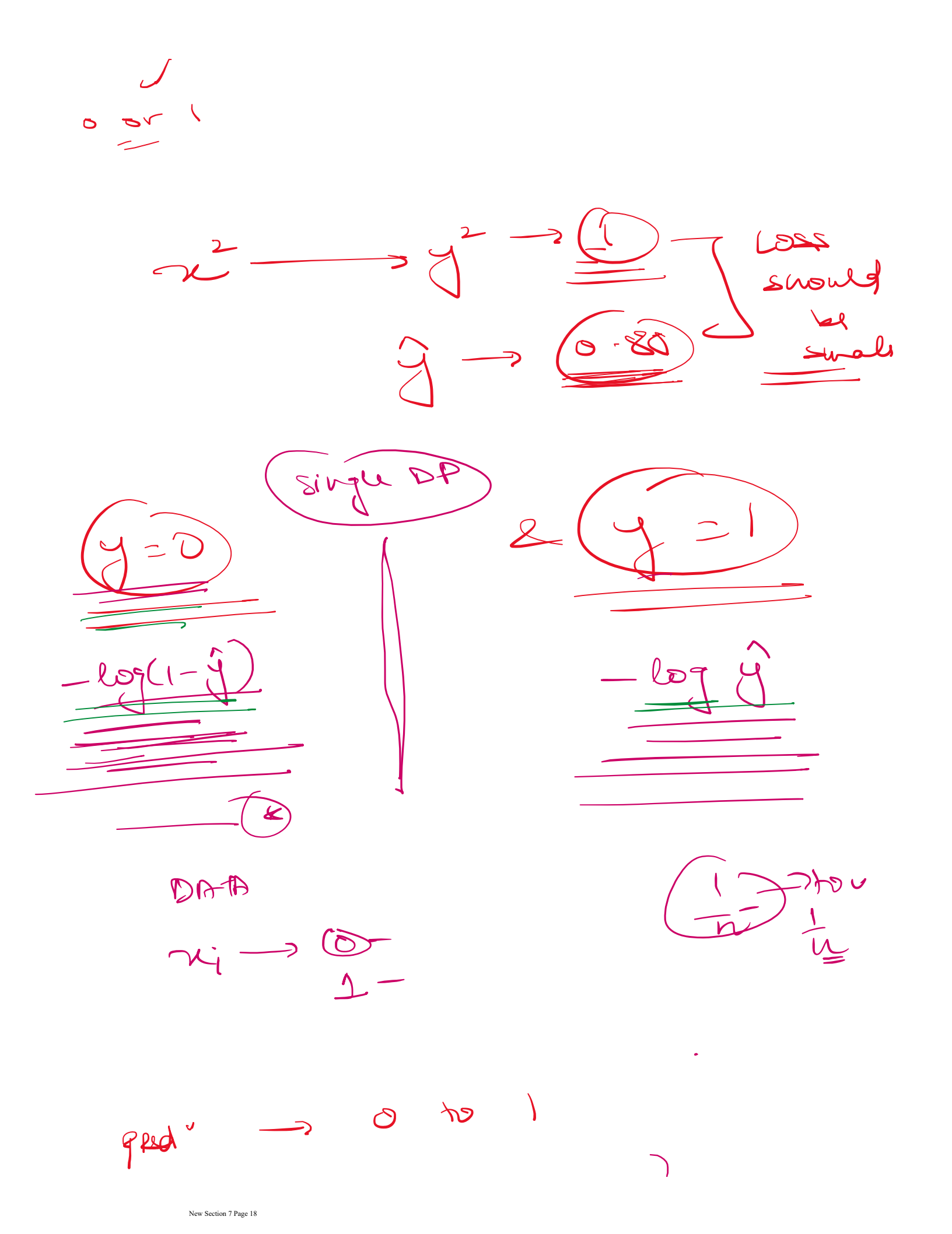
(4) (5) (6)

(2) Log - loss

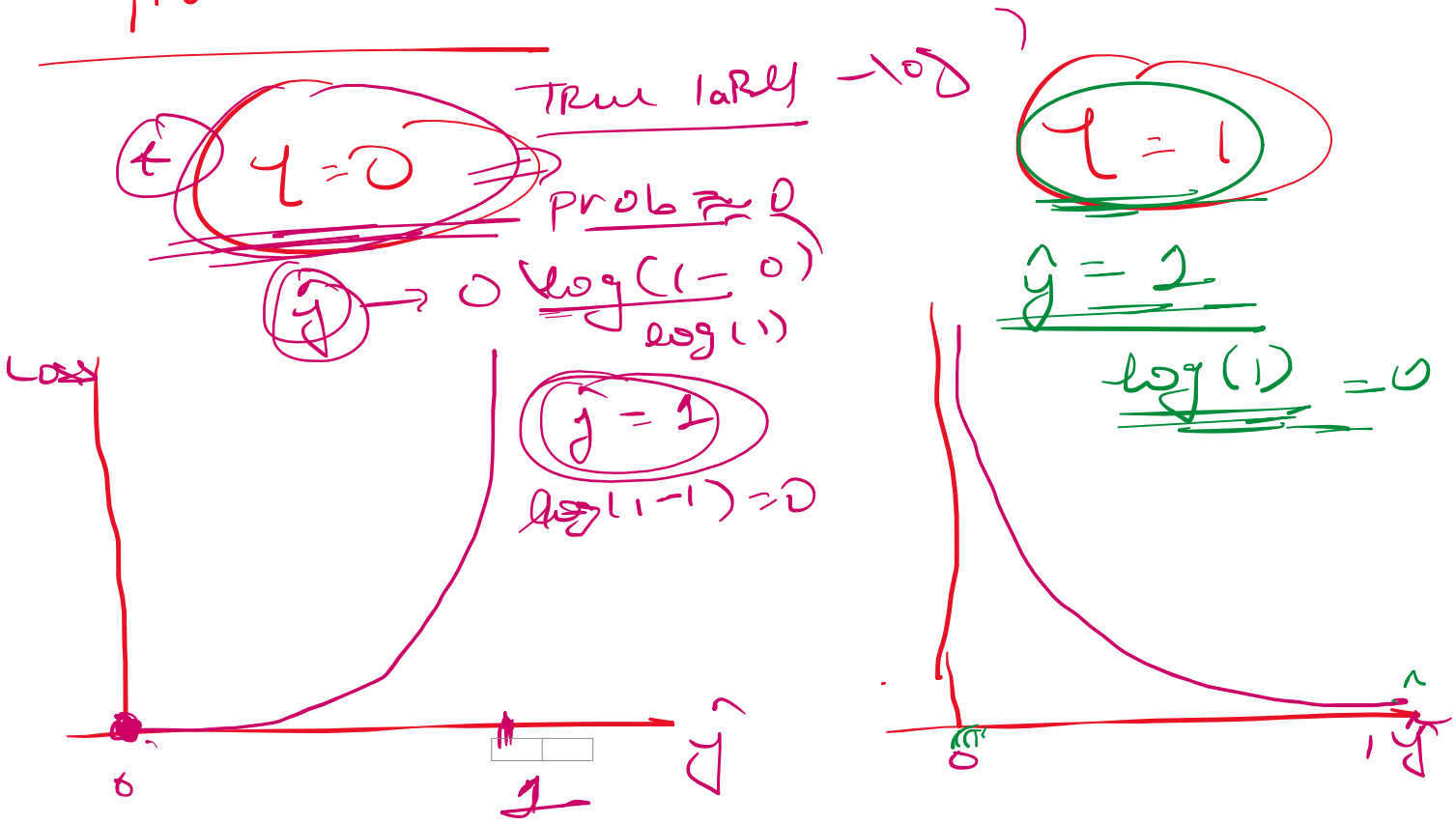
(7) Binary cross entropy (1/2)

$y \in \{0, 1\}$. 0 or 1

$\hat{y} \in (0, 1)$ blw 0 to 1


y
0 or 1
x^2
y^2
1
$\hat{y}$
0.8
Loss should be small
single DP
y=0
y=1
$-\log(1-\hat{y})$
$-\log \hat{y}$
DATA
x_i
0
1
1
$\frac{1}{m}$
grad
0 to 1

Pred ∨ →



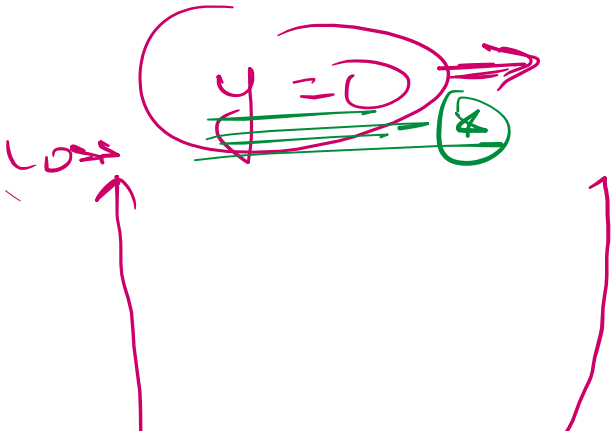
Actual = 0

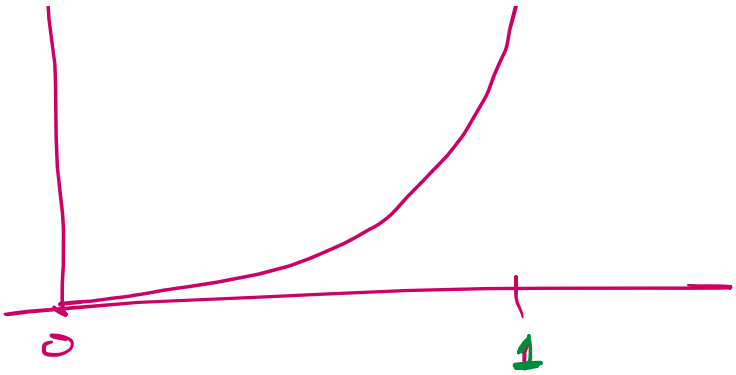
Pred ∨ = 1

= 0

$\log(1-1) = \log(0)$

$= -\infty$





pred^n $\Rightarrow$ (1)

$-\log(1-1) = -\log(0)$

$= -\infty$



True label

$y=0$

$y=1$

good pred

$\hat{y} \approx 0$

$\hat{y} \approx 1$

Bad pred

$\hat{y} \approx 1$

$\hat{y} \approx 0$

Correct pred $\rightarrow$ tiny loss

(4) wrong pred $\rightarrow$ huge loss

Loss func^n

* Optimisation

$$\underset{w, w_0}{\arg\min} \; -\frac{1}{n} \sum_{i=1}^{n} \left[y_i \log(\hat{y}_i) + (1-y_i) \cdot \log(1-\hat{y}_i) \right]$$

eq^n should be minimized

use GD →
(i) init w, w_0
compute gradients
update w_j using

$$w_j = w_j - \eta \frac{\partial L}{\partial w_j}$$

How to find this

chain Rule

→ Next session

MLE

Linear Reg → R^2 R2 Score

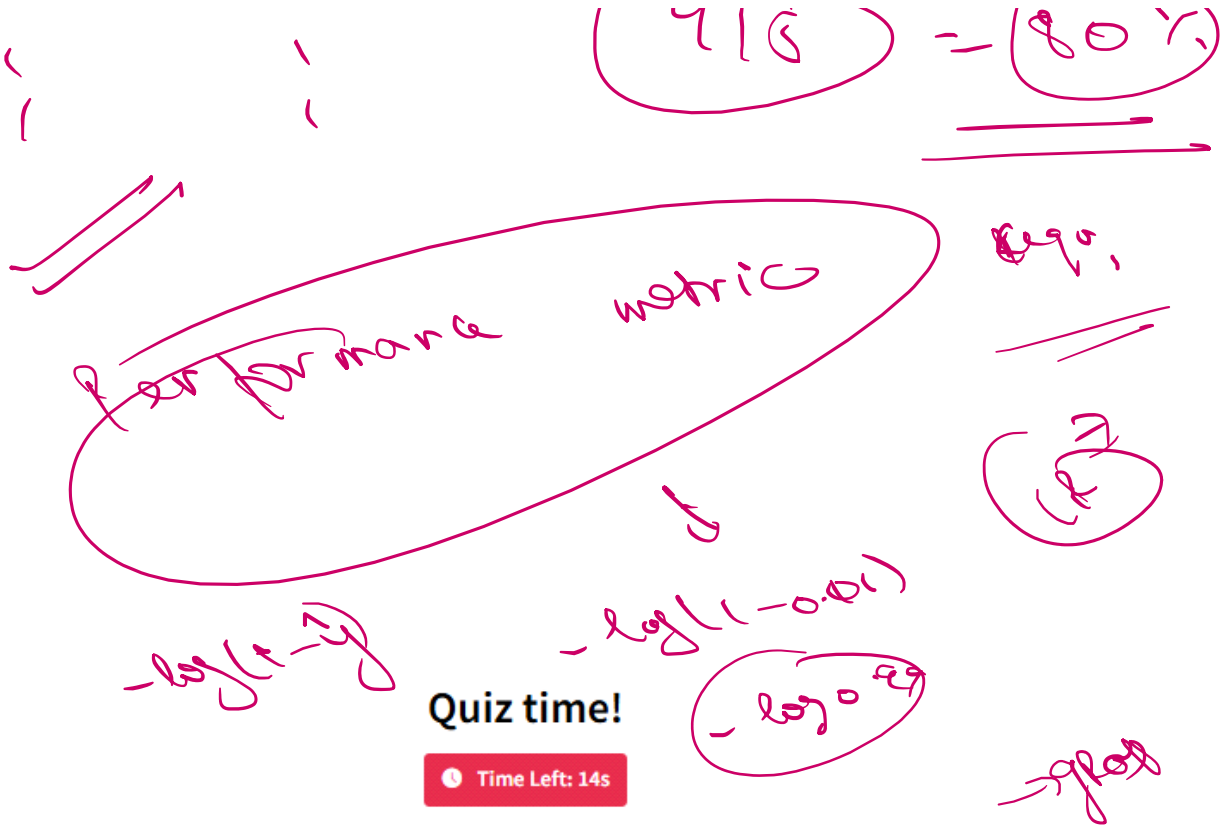
Logistic Reg → accuracy

Pred → correctly

	y	$\hat{y}$
1	1	1
2	0	1
	0	0
	1	1

⇒ accuracy =

4/5 = 80%



Quiz time!

Time Left: 14s

Supposedly your y = 0 and ŷ = 0.01 , so what be the log-loss ?

10 users have participated

A	log-loss will be a very high value	0%
B	log-loss will be a very low value	70%
C	log-loss will be 0	30%

End Quiz Now

